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Emergency department visits associated with wildfire smoke events in California, 2016–2019

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ABSTRACT

Wildfire smoke has been associated with adverse respiratory outcomes, but the impacts of wildfire on other health outcomes and sensitive subpopulations are not fully understood. We examined associations between smoke events and emergency department visits (EDVs) for respiratory, cardiovascular, diabetes, and mental health outcomes in California during the wildfire season June–December 2016–2019. Daily, zip code tabulation area-level wildfire-specific fine particulate matter ($PM_{2.5}$) concentrations were aggregated to air basins. A "smoke event" was defined as an air basin-day with a wildfire-specific $PM_{2.5}$ concentration at or above the 98th percentile across all air basin-days (threshold = 13.5 μg/m^3). We conducted a two-stage time-series analysis using quasi-Poisson regression considering lag effects and random effects meta-analysis. We also conducted analyses stratified by race/ethnicity, age, and sex to assess potential effect modification. Smoke events were associated with an increased risk of EDVs for all respiratory diseases at lag 1 [14.4%, 95% confidence interval (CI): (6.8, 22.5)], asthma at lag 0 [57.1% (44.5, 70.8)], and chronic lower respiratory disease at lag 0 [12.7% (6.2, 19.6)]. We also found positive associations with EDVs for all cardiovascular diseases at lag 10. Mixed results were observed for mental health outcomes. Stratified results revealed potential disparities by race/ethnicity. Short-term exposure to smoke events was associated with increased respiratory and schizophrenia EDVs. Cardiovascular impacts may be delayed compared to respiratory outcomes.

1. Introduction

Wildfires have been increasing in frequency, duration, and severity in the past few decades largely due to climate change, particularly in the western United States (Westerling et al., 2006). In California, warmer temperatures, changes in precipitation regimes, drier conditions, and changes in seasonal prevailing winds have led to the rise of mega-fires and record-setting years of wildfires in recent years. Fifteen of the 20 most destructive wildfires recorded in California in the past century occurred after 2015, with the Camp Fire in 2018 being the most destructive and deadliest, resulting in over 18,000 structures destroyed and 85 deaths (CAL FIRE, 2022). Wildfires have devastating impacts on public health, ecosystems, buildings and infrastructure, and economies, costing the state billions of dollars each year (Feo, T.J. et al., 2020).

Wildfire smoke contains a mixture of particles and gases, with fine particulate matter ($PM_{2.5}$) as the primary pollutant of public health concern due to its toxicity and ability to penetrate deep into the lungs and reach other organs through the bloodstream (Liu et al., 2015; Wildfire Smoke: A Guide for Public Health Officials, 2021). Furthermore, $PM_{2.5}$ can disperse from the source and impact the air quality of communities not directly impacted by the fire. Ambient $PM_{2.5}$ (from multiple sources of emissions) has been shown to be causally linked to cardiovascular disease and strongly linked to other outcomes, including respiratory diseases and nervous system effects (US EPA, 2019). Consistent with the health effects of ambient $PM_{2.5}$, wildfire smoke has been associated with adverse respiratory outcomes in humans and in animals (Black et al., 2017; Carberry et al., 2022; Kim et al., 2018). However, epidemiologic associations between wildfire smoke and other health outcomes, such as cardiovascular diseases, mental health disorders, and adverse birth outcomes, have been mixed (Amjad et al., 2021; Chen et al., 2021; Moore et al., 2006).

There is growing evidence that wildfire-specific $PM_{2.5}$ may be more harmful than ambient $PM_{2.5}$ under non-wildfire conditions. Recent epidemiologic studies have found that wildfire-specific $PM_{2.5}$ may increase the risk of respiratory hospital visits by up to ten times more than that for non-wildfire ambient $PM_{2.5}$ (Aguilera et al., 2021a, 2021b).

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Wildfire-specific $PM_{2.5}$ has also been shown to increase the risk of emergency department visits (EDVs) for headache disorders compared to the risk associated with overall $PM_{2.5}$ (Elser et al., 2023). These associations are supported by a toxicological study that observed an increased risk of inflammation in the lungs of mice after exposure to wildfire smoke compared to the risk after exposure to ambient $PM_{2.5}$ (Wegesser et al., 2009).

Furthermore, there is evidence that historically marginalized communities, including communities of color, are disproportionately exposed to wildfire smoke (Kondo et al., 2022; Méndez et al., 2020; Rappold et al., 2017). Yet, relatively few studies have investigated differential risk of health effects from wildfire smoke based on sociodemographic characteristics (Liu et al., 2017b). A meta-analysis of ten studies on wildfire smoke found a greater impact among females compared to males for asthma and chronic obstructive pulmonary disease and decreased risk of all respiratory outcomes for children compared to adults (Kondo et al., 2019). However, the investigators of the meta-analysis were unable to quantitatively assess effect modification by other sociodemographic characteristics, including race/-ethnicity, due to insufficient numbers.

To address some of these gaps, we investigated associations between short-term exposure to smoke events, derived from wildfire-specific $PM_{2.5}$ concentrations, and EDVs for respiratory, cardiovascular, metabolic, and mental health outcomes in California from 2016 to 2019. We also conducted analyses stratified by race/ethnicity, sex, and age to assess whether some subpopulations are more susceptible to the health effects of smoke events compared to others. These results will contribute to our understanding of the health effects of wildfire smoke and help determine which subpopulations are more vulnerable. Identifying populations at increased risk of wildfire-related health impacts may guide public health officials and other government agencies in developing strategies for reducing wildfire smoke exposures and subsequent health impacts for those who are most affected.

2. Materials and methods

2.1. Exposure data

2.1.1. Wildfire-specific fine particulate matter ($PM_{2.5}$) concentrations

We obtained daily, zip code tabulation area (ZCTA)-level estimates of mean total and wildfire-specific $PM_{2.5}$ concentrations using an ensemble machine learning model (Aguilera et al., 2023). Briefly, three machine learning algorithms were combined to estimate daily, ZCTA-level total $PM_{2.5}$ concentrations, using about 50 predictor variables from satellite data, land use data, and meteorological data. Modeled total $PM_{2.5}$ concentrations showed high correlation with concentrations at all US Environmental Protection Agency's (USEPA) Air Quality System monitoring sites ($R^2 = 0.83$) and with previously published modeled concentrations (Aguilera et al., 2023; Di et al., 2019; Reid et al., 2021). Hazard Mapping System (HMS) smoke plume data were used to identify smoke-exposed zip code days. Counterfactual $PM_{2.5}$ concentrations (i.e. values expected in the absence of wildfire smoke) were estimated by first removing smoke-exposed zip code days from the dataset of total $PM_{2.5}$ concentrations and then imputing non-wildfire $PM_{2.5}$ concentrations using a fast implementation of chained random forest models. Wildfire-specific $PM_{2.5}$ concentrations were obtained by subtracting the counterfactual $PM_{2.5}$ concentrations from the total observed $PM_{2.5}$ concentrations. Although there is no gold standard for wildfire-specific $PM_{2.5}$ concentrations, this dataset was validated by plotting the distribution of wildfire-specific $PM_{2.5}$ concentrations by HMS smoke density: light, medium, and heavy.

California is divided into 15 air basins based on areas with similar geographic and meteorological features for air quality management purposes. Air basins were used as the unit of analysis in this study to obtain sufficient counts of EDVs to assess potential effect modification by race/ethnicity, age, and sex. Since air basins are defined based on similar meteorological and geographic characteristics, the characteristics of regional air pollutants are similar within air basins. Therefore, there is likely to be minimal exposure misclassification of regional air pollutants, such as wildfire-specific $PM_{2.5}$. Daily ZCTA-level concentrations were aggregated to the air basin-level using population-weighted averaging.

2.1.2. Definition of a smoke event

Concentrations of wildfire-specific $PM_{2.5}$ during months January through May were low (maximum concentration = 1.8 $\mu g/m^3$), so we restricted our analysis to months June through December. The distribution of wildfire-specific $PM_{2.5}$ during months June to December was still right-skewed, with a majority of air basin-days having zero exposure to wildfire-specific $PM_{2.5}$. Initial analyses using a continuous variable for wildfire-specific $PM_{2.5}$ revealed minimal associations with health outcomes. We therefore chose to focus on investigating exposure-outcome associations for days with high concentrations of wildfire-specific $PM_{2.5}$. To identify periods of high concentrations of wildfire-specific $PM_{2.5}$, we adapted a previously used exposure variable known as a "smoke wave" (Liu et al., 2017a). Since air basins were used as the unit of analysis, we defined a smoke event as an air basin-day with a wildfire-specific $PM_{2.5}$ concentration at or above the 98th percentile across all air basin-days during June–December 2016–2019 (threshold = 13.5 $\mu g/m^3$).

2.1.3. Wildfire perimeters

We obtained fire perimeters from the California Department of Forestry and Fire Protection's (CAL FIRE) Fire and Resource Assessment Program (https://frap.fire.ca.gov/mapping/gis-data/) (California Department of Forestry and Fire Protection, 2022). This data set consists of fire perimeters for wildfires and prescribed burns contributed by CAL FIRE, the United States Forest Service Region 5, the Bureau of Land Management, and the National Park Service. Fire perimeters were used to create a map showing the number of smoke events for each air basin overlaid with fires that occurred during the study period (Fig. 1).

2.1.4. Co-pollutant data

Monitored data for ozone (O_3), carbon monoxide (CO), and nitrogen dioxide (NO_2) were obtained from the USEPA Air Quality System Data Mart (US EPA, 2020). We included monitors with at least 90% daily coverage during the study period. For each ZCTA, we identified monitors that were within the same air basin and included monitors that were within a maximum distance of 5 km for CO and NO_2 and 10 km for O_3 to the population-weighted centroid. Daily, ZCTA-level concentrations were calculated using inverse distance weighting, based on the monitor distance to the population-weighted centroid. Daily, ZCTA-level concentrations were then aggregated to daily, air basin-level concentrations using population-weighted averaging. The exposure metrics for O_3, CO, and NO_2 were daily 8-h maximum (parts per billion, ppb), daily 1-h maximum (parts per million, ppm), and daily 1-h maximum (ppb), respectively.

2.1.5. Temperature data

Hourly temperature and relative humidity data were obtained from the USEPA, California Irrigation Management Information System (CIMIS), and the National Oceanic and Atmospheric Administration's National Centers for Environmental Information (NCEI) (National Oceanic and Atmospheric Administration, 2021; Office of Water Use Efficiency, CDWR, 2020; US EPA, 2020). We calculated mean daily apparent temperature, also known as the heat index, using formulas as previously described (Basu et al., 2008). Apparent temperature combines air temperature and relative humidity to represent what the temperature feels like.

We included monitors with at least 18 hourly measurements of temperature and relative humidity per day and with at least 90% daily coverage during the study period, December 1, 2015 through December

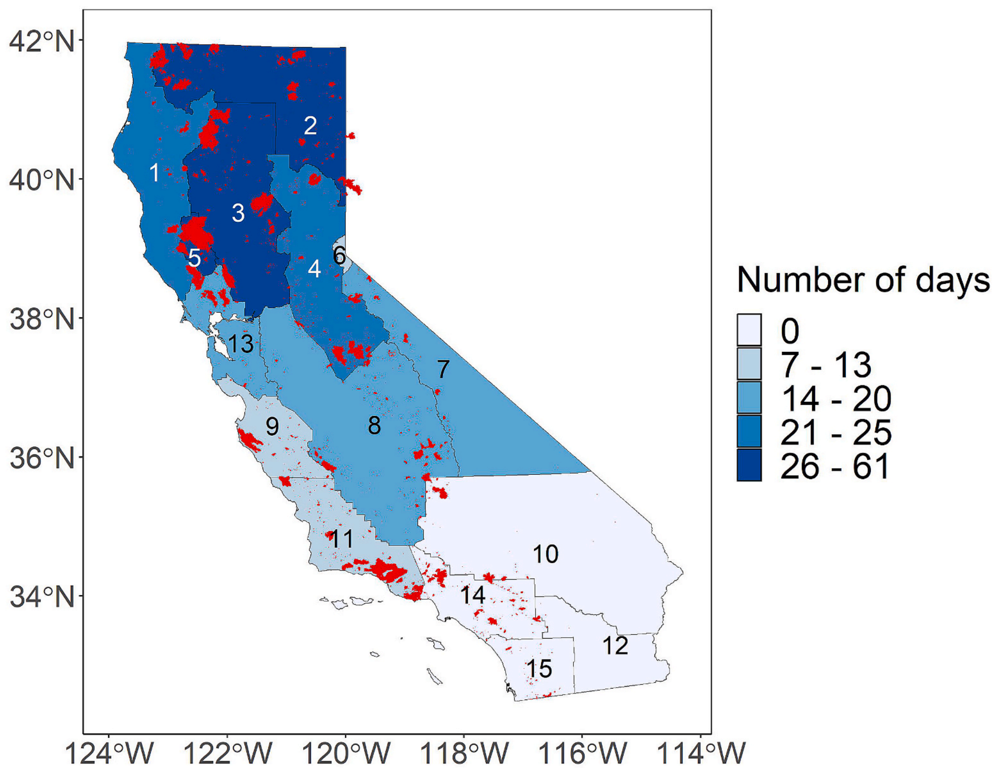

Fig. 1. Map of California indicating the number of smoke events by air basin, 2016-2019 June-December.
Fire perimeters are shown in red. Air basins: 1 – North Coast, 2 – Northeast Plateau, 3 – Sacramento Valley, 4 – Mountain Counties, 5 – Lake County, 6 – Lake Tahoe, 7 – Great Basin Valleys, 8 – San Joaquin Valley, 9 – North Central Coast, 10 – Mojave Desert, 11 – South Central Coast, 12 – Salton Sea, 13 – San Francisco Bay, 14 – South Coast, 15 – San Diego County. The four air basins without any smoke events (i.e. Mojave Desert, Salton Sea, South Coast, and San Diego County) were excluded from analyses.

31, 2019 (including one month before the start of the study period). For each ZCTA, we identified monitors that were within the same climate zone and air basin. The mean daily apparent temperature values for each ZCTA were calculated using inverse distance weighting, based on the monitor distance to the population-weighted centroid. Daily, ZCTA-level values were aggregated to daily, air basin-level values using population-weighted averaging.

2.2. Health outcome data

We prepared daily counts of unscheduled EDVs for years 2016–2019 using the Emergency Department Data and Patient Discharge Data datasets provided by the California Department of Health Care Access and Information (California Department of Health Care Access and Information (HCAI), 2021). The Patient Discharge Data were used to identify unscheduled acute or psychiatric inpatient hospitalizations that originated from the hospital's emergency department. We chose to include psychiatric care to fully capture EDVs for mental health disorders. For each encounter, we extracted the following variables: service or admission date, *International Classification of Disease, Tenth Revision, Clinical Modification* (ICD-10-CM) code for the principal diagnosis, race/ethnicity, age, sex, and residential zip code. Daily counts were summed by ZCTA and aggregated to the air basin level.

The principal diagnosis, reported using ICD-10-CM diagnosis codes, was used to group case counts into outcome categories for respiratory, cardiovascular, diabetes, and mental health outcomes. We used the Agency for Healthcare Research and Quality's (AHRQ) Clinical Classifications Software Refined (CCSR) tool, developed as part of the Healthcare Cost and Utilization Project, to group EDVs into clinically meaningful and mutually exclusive diagnosis categories ("Clinical Classifications Software Refined (CCSR) for ICD-10-CM Diagnoses," 2022). Version 2022-1 of the SAS mapping program provided by AHRQ (DXCCSR_Mapping_Program_v2022-1.sas) was used to map ICD-10-CM codes to their default CCSR category, based on principal diagnosis for inpatient data and the first-listed diagnosis for outpatient data. Version 2022-1 is compatible with ICD-10-CM codes from October 2015 through September 2022, which captures our entire study period. We reviewed the DXCCSR-Reference-File-v2022-1 to determine which default CCSR categories to group together for our final outcomes of interest and made manual adjustments for specific ICD-10-CM codes as needed (Table 1, Table S1). The final outcome categories included in this study were: all respiratory diseases, asthma, acute upper respiratory infections (AURI), chronic lower respiratory disease (CLRD), pneumonia, all cardiovascular diseases, acute myocardial infarction (AMI), cerebrovascular disease, dysrhythmia, ischemic heart disease (IHD), peripheral vascular disease (PVD), transient ischemic attack (TIA), diabetes, all mental health disorders, anxiety, mood disorders, schizophrenia, substance use disorders, and suicide/intentional self-harm (suicide) (Table 1).

Outcome counts were also stratified by race/ethnicity, age, and sex for subgroup analyses. The race/ethnicity categories were: Hispanic, non-Hispanic White ("White"), non-Hispanic Black ("Black"), and non-Hispanic Asian and Pacific Islander ("API"). Prior to 2019, the categories Asian and Native Hawaiian and Pacific Islander were combined into one category, so we combined those two categories for the data in 2019 to ensure consistency in the race/ethnicity coding. The age groups were as follows: 0–17 years old, 18–64 years old, and 65+ years old. For sex, we included males and females.

2.3. Population data

ZCTA-level population counts and sociodemographic data from the 2015–2019 5-year American Community Survey (ACS) were obtained using the tidycensus package in R and were used to calculate the total population and stratified population counts for each air basin (Walker and Herman, 2023). We used 2010 ZCTA population-weighted centroids and conducted a spatial join to assign each ZCTA to an air basin.

2.4. Statistical analysis

2.4.1. Main model

We conducted a two-stage time-series analysis to investigate associations between smoke events and EDVs for various causes. In the first

Table 1
Outcome categories and their corresponding Clinical Classifications Software Refined (CCSR) codes.

Outcome Group	CCSR codes
All respiratory diseases	RSP001-RSP017
Asthma	RSP009
Acute upper respiratory infections (AURI)	RSP001, RSP004, RSP006
Chronic lower respiratory disease (CLRD)	RSP008
Pneumonia	RSP002
All cardiovascular diseases	CIR001-CIR036, CIR038-CIR039
Acute myocardial infarction (AMI)	CIR009
Cerebrovascular disease	CIR020-CIR025
Dysrhythmia	CIR016-CIR018
Ischemic heart disease (IHD)	CIR009-CIR011
Peripheral vascular disease (PVD)	CIR026-CIR027, CIR029-CIR030, CIR032-CIR036, CIR039
Transient ischemic attack (TIA)	NVS012
Diabetes	END002-END003
All mental health disorders	MBD001-MBD014, MBD017-MBD027, MBD034
Anxiety	MBD005-MBD007, MBD011
Mood disorders	MBD002-MBD004, select ICD-10-CM codes from MBD026
Schizophrenia	MBD001
Substance use disorders	MBD017-MBD025, select ICD-10-CM codes in MBD026 and MBD034
Suicide	MBD012, MBD027, select ICD-10-CM codes in MBD034 and INJ075

stage, we ran time-series models for each air basin using a quasi-Poisson regression to account for overdispersion. In the second stage, we conducted random effects meta-analyses to combine air basin-specific estimates into an overall relative risk (RR) and corresponding 95% confidence intervals (CI), using the mixmeta package in R (Sera et al., 2019). The results are shown as percent change in risk, which was calculated as (RR – 1)*100%. To account for confounding, we adjusted for seasonal/long-term trends, holidays, day of the week, and mean apparent temperature at lag 0. We used lag 0 for mean apparent temperature because previous studies in California have reported positive associations between same-day mean apparent temperature and EDVs for various health outcomes (Basu et al., 2012, 2018). A natural cubic spline smoothing function of time was used to adjust for seasonal/long-term trends. The following holidays were combined into a single binary indicator variable: New Year's Eve, New Year's Day, Memorial Day, July 4th, Labor Day, Thanksgiving, and Christmas Day. We used six indicator variables for days of the week, with Sunday as the reference. We also included the log of the population as an offset to model rates rather than counts of EDVs, since population sizes varied across air basins and demographic groups. The main model was specified as follows:

$$Y_{ij} \sim \ Poisson(\mu_{ij})$$

$$\log(\mu_{ij}) = \alpha + \beta_1(smoke\ event\ indicator_{ij}) + \beta_2(mean\ apparent\ temp_{ij}) + \beta_3(day\ of\ week_j) + \beta_4(holiday_j) + ns\left(date_j,\ df = number\ of\ years * \frac{df}{yr}\right) + \log(population_i)$$

where Y = EDV counts, i = air basin, j = observation date, df/yr = degrees of freedom per year.

Final model specifications were chosen by maximizing model fit. We chose the model with the smallest sum of the quasi-Akaike Information Criterion (qAIC) across the first-stage quasi-Poisson regression models. For the natural cubic spline, we tested between 1 and 6 degrees of freedom per year (df/yr). Four df/yr resulted in the best model fit for respiratory outcomes; for all other outcomes, we used two df/yr because of better model fit.

We also tested multiple lags and used the lag with the lowest total qAIC for each outcome. We looked at single-day lags and cumulative lags to investigate the effect of consecutive days of smoke events. For cumulative lags, each day in the lag period had to meet the threshold for a smoke event for the period to be considered a smoke event. We considered the following lags: same day (lag 0), previous day (lag 1), two days prior (lag 2), and cumulative lags of two days (lag 0_1) and three days (lag 0_2). Since previous studies have shown adverse respiratory outcomes within the first few days following wildfire smoke exposure, we did not test lags past 2 days for respiratory outcomes (Delfino et al., 2009; Reid et al., 2016b). For all other outcomes, we also considered longer lags: one week prior (lag 7), ten days prior (lag 10), and two weeks prior (lag 14). We used lag 0 for mean daily apparent temperature for all models.

We also assessed potential confounding by other air pollutants, including non-wildfire $PM_{2.5}$, O_3, CO, and NO_2. Some air basins were excluded from this analysis, due to insufficient monitor coverage across the state. For each air pollutant, we used the same subset of data and ran the main model as specified above with and without the co-pollutant in the model. Associations between smoke events and EDVs with and without the co-pollutant in the model were compared to determine whether there was potential confounding. We used the same lag for the co-pollutant and the main smoke event indicator for each outcome. Since these additional air pollutants are not thought to be associated with the concentration of wildfire-specific $PM_{2.5}$, we did not include these variables in the main model. In addition, we wanted to use the same set of covariates across all the health outcomes and opted to assess the robustness of the results by conducting sensitivity analyses with and without additional air pollutants in the model.

2.4.2. Effect modification

We conducted stratified analyses to assess potential effect modification by race/ethnicity, age group, and sex. We conducted a Z test of interaction to assess differences among subgroups, using Whites, 18–64 years old, and males as the reference groups, respectively (Altman, 2003). Some demographic and outcome combinations were excluded from stratified analyses due to insufficient numbers.

2.4.3. Sensitivity analyses

To assess the robustness of our results to different smoke event definitions, we used air basin-specific 98th percentile thresholds to identify smoke events for each air basin. We then compared associations based on the overall threshold, basin-specific thresholds for all 15 air basins, and basin-specific thresholds using the same subset of 11 basins as in the main analysis.

2.4.4. Health impact function

We applied a log-linear health impact function to estimate the number of EDVs attributable to smoke events. The effect estimate for the smoke event indicator was obtained from the second-stage meta-analysis that combined the air basin-specific estimates into an overall estimate for a given outcome (Table S2). For each outcome, attributable EDVs were calculated for each air basin separately, using air basin-specific values for baseline incidence rate, population, and number of smoke event days. Outcome-specific daily baseline incidence rates were calculated for each air basin by dividing the total number of EDVs for that outcome in a given air basin by the number of days in the study period (214 days/year * 4 years) and then dividing by the air basin population. Air basin-specific attributable EDVs were summed to obtain the total attributable EDVs on smoke event days for the 11 air basins included in this study. The following equation describes the health impact function used to estimate health burden on smoke event days:

$$\Delta y_{SEj} = \sum_{i}^{AB(n=11)} y_{0ij} * \left[\exp\left(\beta_{SEj}\right) - 1\right] * Population_i * SmokeEventDays_i$$

where i = air basin, j = outcome group, β_{SE} = overall effect estimate for a smoke event day, y_0 = baseline incidence rate of EDVs, SmokeEventDays = number of smoke event days during the study period, and Δy_{SE} = total number of EDVs attributable to smoke events. It should be noted that β_{SE} is the effect estimate for the binary smoke event indicator.

Outcome data were cleaned and prepared using SAS version 9.4. All other analyses were conducted in RStudio using R version 4.2.1 (R Core Team, 2021; RStudio Team, 2020).

3. Results

3.1. Descriptive statistics of exposure data

Across the 15 air basins in California, daily wildfire-specific $PM_{2.5}$ concentrations from June–December 2016–2019 ranged from 0 to 164.6 μg/m^3 (median = 0 μg/m^3, mean = 1.1 μg/m^3), with 95% of air basin-days having concentrations below 5 μg/m^3. There were 257 smoke events out of 12,840 air basin-days. Four of the air basins (South Coast, San Diego, Salton Sea, and Mojave Desert) were excluded from this study, since they did not have any smoke events during the study period, based on our definition (Fig. 1).

Among the remaining 11 air basins, the median wildfire-specific $PM_{2.5}$ concentration for smoke events was 23.1 μg/m^3 (interquartile range (IQR): 14.9), compared to a median of 0 μg/m^3 (IQR: 0.2) on days that were not smoke events (non-smoke events). The mean number of smoke events per basin was 23.4. The air basins with the fewest and greatest number of smoke events were North Central Coast (n = 7) and the Northeast Plateau (n = 61), respectively (Fig. 1).

3.2. Description of study population and outcome data

The 11 air basins with smoke events during the study period included 17,847,917 people, which is about 45.4% of California's population. The study population had a similar age group and sex distribution to that of the entire state (Table 2). With respect to race/ethnicity composition, the study population had a slightly higher percentage of people who were White (42.6% vs 37.2%) or API (16.2% vs 14.6%) and a lower percentage of people who were Hispanic (32.1% vs 39.0%), compared to the breakdown for the entire state.

The number of EDVs totaled 1,363,054 for all respiratory diseases, 1,166,893 for all cardiovascular diseases, 146,888 for diabetes, and 630,113 for all mental health outcomes (Table 3). The most common disease subgroups for respiratory diseases, cardiovascular diseases, and mental health disorders were AURI (45.7% of all respiratory EDVs), dysrhythmia (10.7% of all cardiovascular EDVs), and substance use disorders (36.1% of mental health EDVs), respectively. The mean daily rate of EDVs per 100,000 people was higher for smoke events compared to non-smoke events for most respiratory disease categories in this study. For all other outcomes, the rate of EDVs was similar between smoke events vs non-smoke events.

3.3. Associations between smoke events and EDVs

Smoke events were associated with EDVs for several disease outcomes, with the greatest associations observed for respiratory outcomes (Fig. 2, Fig. S1, Table S2). The lag that resulted in the best model fit differed by disease subgroup, but results were generally consistent across lags for most outcomes. Smoke events were associated with a 14.3% (95% CI: 6.8, 22.5%) increase in the risk of EDVs for all respiratory diseases at lag 1. Smoke events showed positive associations with all respiratory disease subgroups, with the strongest increase in the risk for asthma EDVs at lag 0 (57.1%, 95% CI: 44.5, 70.8%). EDVs for pneumonia and AURI showed positive associations with smoke events, although they were not statistically significant (Table S2).

Smoke events were also associated with a 3.2% (95% CI: 1.7, 4.7%) increase in the risk of EDVs for all cardiovascular diseases at lag 10. The association between smoke events and EDVs for all cardiovascular diseases was negative or null during earlier lags but gradually became positive with longer lags, suggesting that cardiovascular impacts may be delayed compared to respiratory impacts. No significant associations for specific cardiovascular disease subgroups were observed (Fig. S1). The association between smoke events and diabetes EDVs was positive, but not statistically significant. With regards to mental health EDVs, smoke events were positively associated with EDVs for schizophrenia at lag 0 (13.1% increase in risk, 95% CI: 5.1, 21.8%) but significantly negatively associated with EDVs for substance use disorders and suicide (Fig. 2, Fig. S1, Table S2). The associations for schizophrenia and suicide were consistent across short-term lags through day 2, but changed directions with longer lags tested. The negative association for substance use disorders was observed in other lags tested, but the association was only statistically significant for lag 10, suggesting that this result may be more sensitive to the specific lag (Fig. 2).

Our results were generally robust to adjustment for co-pollutants (Fig. S2-S5). The models with and without non-wildfire $PM_{2.5}$ included 11 air basins, the same dataset as in the main model. Analyses for O_3, NO_2, and CO included fewer air basins (eight, five, and six, respectively) due to insufficient monitor coverage across the state. Adjustment for CO or non-wildfire $PM_{2.5}$ resulted in greater reductions in the magnitude of associations, particularly for respiratory outcomes, but the results were still similar to unadjusted estimates (Fig. S2, S5). Given the fewer number of air basins included in some co-pollutant analyses, the results may not be as representative of the entire state.

We also assessed the robustness of our results to different smoke event definitions by comparing results based on the overall threshold to those using basin-specific thresholds. For the same subset of 11 air basins, we found similar results between these two threshold definitions (Fig. S6). For some outcomes, namely pneumonia, mood disorders, and schizophrenia, results based on basin-specific thresholds led to a slightly higher increase in the risk of EDVs. We also observed similar results based on basin-specific thresholds with and without the four basins excluded from the main analysis, although the increase in the risk of asthma EDVs was lower when all basins were included (Fig. S6). This observation was likely driven by the four basins that were excluded in the main analysis, since the first-stage model estimates for asthma EDVs for these four basins were close to null (Fig. S7).

Using a health impact function, we calculated the health burden attributable to smoke events, based on the relative risk estimates we

Table 2
Demographic characteristics of the entire state and the study population.

Demographic group	Entire state	Study population
N	**39,275,325**	**17,847,917**
Age group (years) (%)		
0-17	22.97	23.04
18-64	63.06	62.31
65+	13.97	14.66
Race/ethnicity (%)		
White	37.18	42.62
Black	5.52	4.71
Hispanic	39.02	32.12
API	14.64	16.25
Other	3.64	4.30
Sex (%)		
Male	49.70	49.81
Female	50.30	50.19

The Other subgroup was not included in our analysis stratified by race/ethnicity.

Table 3
Mean daily rate of EDVs per 100,000 people for each outcome on all days, smoke event days, and non-smoke event days.

Outcome Group	N	Mean daily rate of EDVs per 100,000 people		
		All days	Smoke event days	Non-smoke event days
All respiratory diseases	1,363,054	8.92	10.03	8.90
Asthma	163,994	1.07	1.77	1.06
AURI	622,512	4.07	4.40	4.07
CLRD	102,189	0.67	0.74	0.67
Pneumonia	143,820	0.94	0.92	0.94
All cardiovascular diseases	1,166,893	7.64	7.55	7.64
AMI	59,686	0.39	0.41	0.39
Cerebrovascular disease	83,924	0.55	0.54	0.55
Dysrhythmia	124,935	0.82	0.82	0.82
IHD	86,699	0.57	0.59	0.57
PVD	67,111	0.44	0.44	0.44
TIA	27,261	0.18	0.17	0.18
Diabetes	146,888	0.96	0.99	0.96
All mental health disorders	630,113	4.12	4.03	4.13
Anxiety	162,732	1.07	1.03	1.07
Mood disorders	72,896	0.48	0.46	0.48
Schizophrenia	66,577	0.44	0.46	0.44
Substance use disorders	227,765	1.49	1.44	1.49
Suicide	83,584	0.55	0.52	0.55

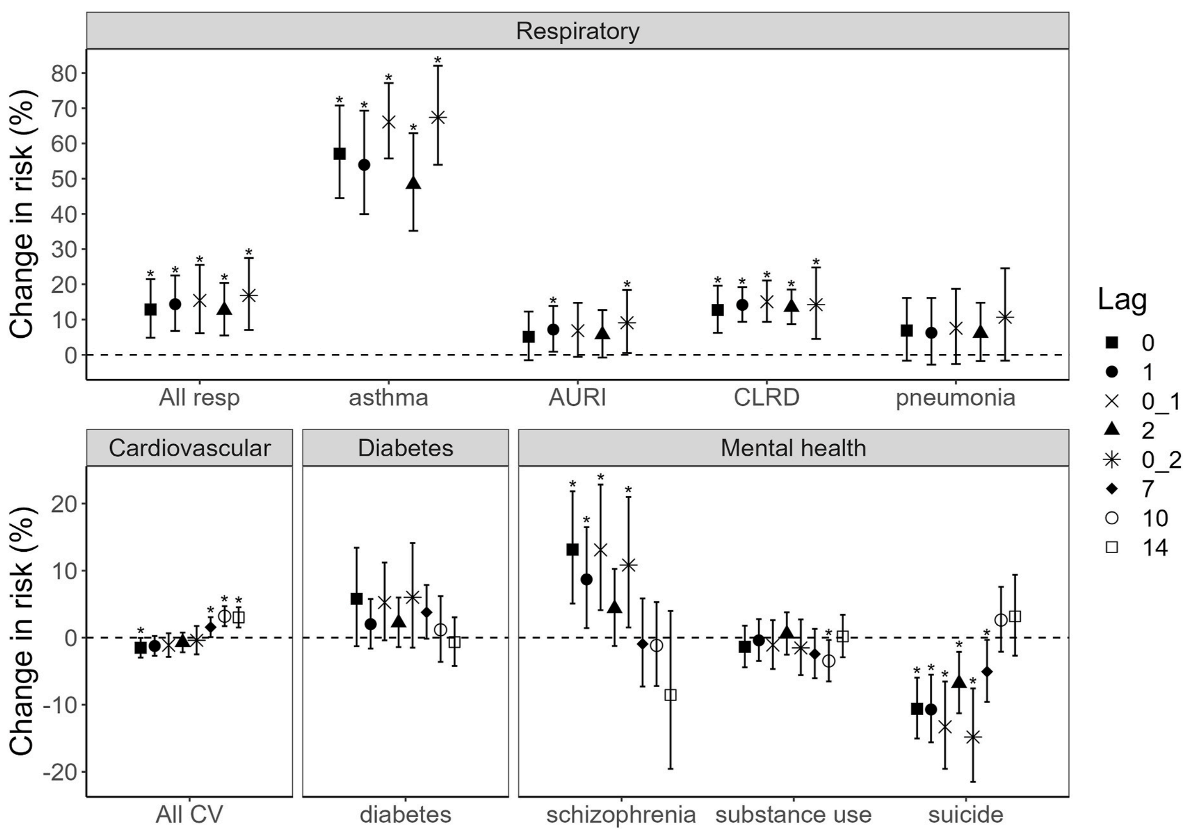

Fig. 2. Percent change in risk of EDVs for select respiratory, cardiovascular, diabetes, and mental health diagnoses associated with smoke events. The lag with the best model fit is as follows: All resp – lag 1, asthma – lag 0, AURI – lag 2, CLRD – lag 0, pneumonia – lag 0_2, All CV – lag 10, diabetes – lag 0_2, schizophrenia – lag 0, substance use – lag 10, and suicide – lag 0_1. All resp: all respiratory diseases; All CV: all cardiovascular diseases. Bars: 95% confidence intervals. Dotted line represents no change in risk. *$p < 0.05$.

obtained (Table S2). The estimated health burden attributable to smoke events was greatest for all respiratory diseases combined, with 4597 (95% CI: 2164, 7203) attributable EDVs. There were 2204 (95% CI: 1718, 2733) attributable EDVs for asthma and 323 (95% CI: 158, 498) attributable EDVs for CLRD. We estimated 889 (95% CI: 476, 1307) attributable EDVs for all cardiovascular diseases combined. There were an estimated 208 (95% CI: 81, 345) attributable EDVs for schizophrenia.

3.4. Subpopulations at increased risk of EDVs associated with smoke events

Stratification analyses by race/ethnicity identified a few statistically significant differences among subpopulations (Fig. 3, Table S3). The risk of diabetes EDVs at lag 0_2 for people who are Hispanic (13.7%, 95% CI: 2.2, 26.4%) was significantly higher than the risk for people who are White (−2.6%, 95% CI: −9.8, 5.2%) (Fig. 3A, Table S3). The risk of diabetes EDVs was trending positive among people who are Black or API, but we did not observe a significant difference compared to the risk for people who are White. For anxiety, we observed a significantly lower risk of EDVs among people who are API compared to people who are White, although neither subgroup had a significant change in risk compared to the null (Fig. 3B, Table S3).

There were also some potential differences that may be worth further investigation in future studies. While there were no significant differences in the risk of asthma EDVs across race/ethnicity subgroups, the risk was trending lower among people who are Black or API (Fig. 3C). For pneumonia, we observed the greatest increase in risk of EDVs for people who are API or Hispanic, although the risk was elevated for all race/ethnicity subgroups (Fig. 3D). There were no significant differences in the risk of EDVs for cardiovascular disease outcomes among race/ethnicity subpopulations (Table S3). However, the risks of EDVs for AMI and dysrhythmia were trending positive for people who are Black (Fig. 3E and F, Table S3).

There were some differential effects by age. People ages 65 and older had a significantly higher risk of EDVs for transient ischemic attacks at lag 2 compared to people ages 18–64 years old, although the confidence interval contained the null value (8.3%, 95% CI: −7.9, 27.4%) (Table S3). Children ages 0–17 years had a significantly lower risk of EDVs for all mental health outcomes at lag 7 (−12.7%, 95% CI: −18.6, −6.4%), compared to the risk for people ages 18–64 years old.

There were no statistically significant differences by sex, but the increases in risk for asthma and CLRD were trending higher among females.

4. Discussion

In this study, we investigated short-term associations between smoke events and EDVs for various causes in California from June–December 2016–2019. Our results are consistent with previous epidemiologic studies in California that found positive associations between wildfire smoke and short-term adverse respiratory effects, most notably for asthma and CLRD (Aguilera et al., 2021a; Delfino et al., 2009; Heaney et al., 2022; Reid et al., 2016b). While it is difficult to quantitatively compare estimates across studies, given study differences in smoke event thresholds, time periods, and geography, the magnitude of associations we observed for respiratory EDVs (~14% increase) is reasonable, compared to those reported in previous studies. For example, a study during 2011–2017 in San Diego, California found a 30% increase in respiratory EDVs for a 10-unit increase in wildfire-specific $PM_{2.5}$ (Aguilera et al., 2021a), whereas two other studies found a ~3% increase in respiratory EDVs during the 2008 northern California wildfires and for smoke event days during the 2004–2009 wildfire seasons (Heaney et al., 2022; Reid et al., 2016b). We also observed positive associations between smoke events and EDVs for all cardiovascular diseases combined at a later lag but did not find any associations for specific cardiovascular disease subgroups, which is largely consistent with the findings of a recent review (Chen et al., 2021). Our finding of a positive association between smoke events and schizophrenia EDVs also adds to our understanding of mental health effects of wildfire smoke (Eisenman and Galway, 2022; To et al., 2021).

Our observation of strong, immediate respiratory effects and more delayed cardiovascular effects may be partly explained by the timing of the physiological effects of wildfire smoke. Toxicological studies have shown that animals exposed to wildfire smoke initially experience increased inflammation in the lungs, which is followed by inflammation and tissue injury in other organs such as the heart (Black et al., 2017; Kim et al., 2014, 2018). Furthermore, a recent study of mice exposed to wildfire smoke reported changes in the expression of genes involved in hypoxia and cell stress signaling pathways, which have been shown to be involved in tissue injury and inflammation in the heart (Carberry et al., 2022; Giussani et al., 2014). These gene expression changes were first observed in the lungs and were followed by the release of plasma extracellular vesicles (EVs) containing microRNAs (miRNAs) involved in those pathways and changes in expression of the same miRNAs in the heart (Carberry et al., 2022). The earlier effects observed in the lungs are consistent with the strong positive association between smoke events and short-term respiratory impacts that we found. These toxicological findings are also consistent with our finding of a positive association between smoke events and EDVs for cardiovascular diseases at a later lag (lag 10), although we did not detect significant associations with EDVs for any cardiovascular disease subgroups.

Several biological mechanisms of action have been proposed for the cardiorespiratory effects of wildfire smoke. Wildfire smoke inhaled into the lungs is thought to activate neural receptors, leading to changes in

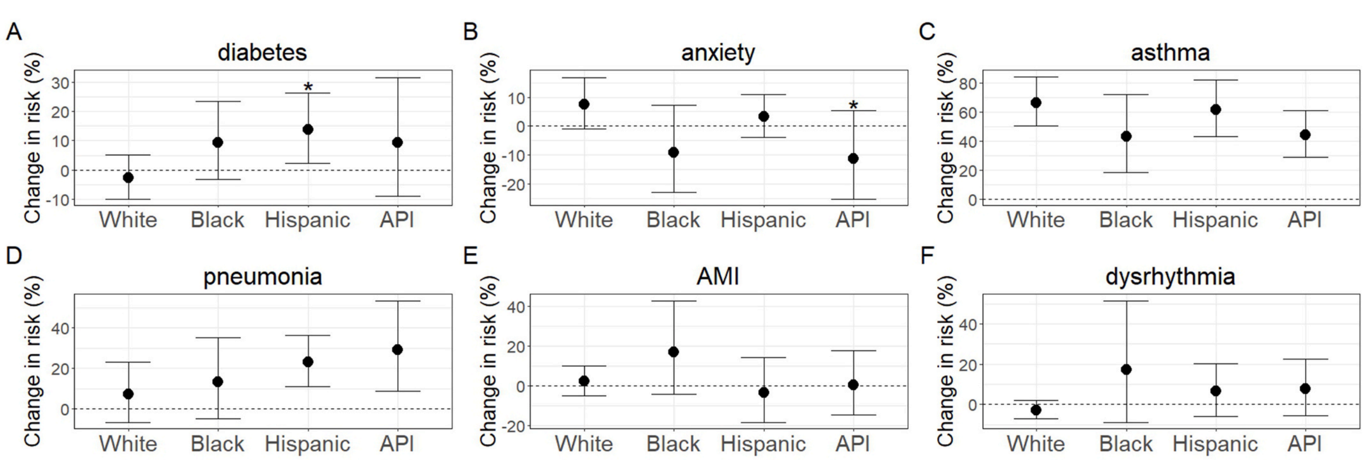

Fig. 3. Percent change in risk of EDVs for (a) diabetes, (b) anxiety, (c) asthma, (d) pneumonia, (e) AMI, and (f) dysrhythmia associated with smoke events, stratified by race/ethnicity.

The lag with the best model fit for each outcome is shown: diabetes – lag 0_2, anxiety – lag 0_1, asthma – lag 0, pneumonia – lag 0_2, AMI – lag 0, and dysrhythmia – lag 1. Bars: 95% confidence intervals. Dotted line represents no change in risk. *$p < 0.05$ for the Z test of interaction..

heart rate and blood pressure (Cascio et al., 2018). Wildfire smoke in the lung has also been shown to increase oxidative stress and systemic inflammation, leading to disruptions in the cell cycle and release of EVs, which play important roles in cell-cell communication. EVs transport molecules including miRNAs, which can alter expression of genes and proteins in target cells (Cascio et al., 2018; Chen et al., 2021). Finally, ultrafine components of wildfire smoke may directly translocate into the bloodstream and reach other organ systems through the circulatory system. Studies have shown that the chemical composition of $PM_{2.5}$ from wildfire smoke has a higher proportion of organic compounds compared to non-wildfire $PM_{2.5}$ (Liu and Peng, 2019; Makkonen et al., 2010; Verma et al., 2009). In addition, smoke-impacted days due to fires in the wildland-urban interface have been observed to have higher concentrations of trace metals and other chemicals harmful to human health, with distinct chemical profiles for particular wildfires (Boaggio et al., 2022; California Air Resources Board, 2021). Future studies on the health effects of wildfire-specific $PM_{2.5}$ constituents may help elucidate the toxicity of wildfire smoke and how this may differ depending on the materials burned.

Associations between smoke events and EDVs for mental health disorders were mixed. We observed positive associations between smoke events and EDVs for schizophrenia and negative associations for EDVs for most of the other mental health outcomes included in this study. Schizophrenia is a mental illness characterized by hallucinations, delusions, and disordered thinking and behavior. One of the disease phenotypes of schizophrenia involves neuroinflammation and alterations to the immune system, which has been observed in mice that were exposed to wildfire smoke-derived particulate matter (Comer et al., 2020; Kroken et al., 2018; Müller and Schwarz, 2010; Scieszka et al., 2022). Other factors that may have played a role include increased psychosocial stress during smoke events, reduced ability to relocate or otherwise reduce smoke exposure, and factors related to medication use, including medication nonadherence that may have exacerbated psychotic symptoms or adverse side effects of antipsychotic medications (Block and Calderón-Garcidueñas, 2009; Bretler et al., 2019; Haddad et al., 2014; McCann et al., 2009). While there have been mixed results from epidemiologic studies on the association between ambient, non-wildfire $PM_{2.5}$ and schizophrenia (Gao et al., 2017; Newbury et al., 2019; Nguyen et al., 2021; Wei et al., 2022), our results indicate that the association between wildfire smoke and schizophrenia is worth further investigation.

For other mental health disorders, our findings were consistent with previous studies using data on hospital visits that found null or negative associations between wildfire smoke and short-term mental health effects (Malig et al., 2021; Moore et al., 2006; Reid et al., 2016a). There are several potential explanations for this finding. The presence of wildfires may have reduced social activities, and consequently, EDVs related to substance use. In addition, patients may have chosen to avoid coming to the hospital for less urgent or non-life-threatening conditions during the initial days following smoke events (Schranz et al., 2010). Hospitals may have also prioritized treating patients for more urgent issues, such as respiratory outcomes. This is consistent with our observation of lower rates of EDVs for most mental health outcomes on days that were smoke events compared to days that were not smoke events (Table 3). Mental health effects also could have exacerbated pre-existing health conditions and contributed to increased visits for respiratory effects or other physical diseases. For example, a survey of a medically vulnerable population in Mariposa County, California found that a respondent's anxiety from the presence of smoke contributed to their cardiovascular symptoms (Hoshiko et al., 2023). It is possible that patients would seek care primarily for acute, physical symptoms rather than potential underlying mental health issues. Qualitative studies may be more suited for understanding the mental health impacts of wildfires, as they extend beyond diagnostic codes and can provide much richer information on how people's mental health and well-being may be affected (Brown et al., 2021; Isaac et al., 2023; Silveira et al., 2021; To et al., 2021). In addition, long-term exposure may be a more appropriate metric, given the diverse pathways through which wildfire smoke may act, including reduced physical activity, social isolation, uncertainty due to lack of access to timely information, and impacts of evacuation and/or livelihood (Eisenman et al., 2021; Eisenman and Galway, 2022).

Our stratified analyses identified significant effect modification by race/ethnicity or age for some outcomes. We observed a higher risk of EDVs for diabetes among people who are Hispanic and a lower risk of EDVs for anxiety among people who are API. However, we did not observe differences in the risk of EDVs for respiratory outcomes between children and adults, as has been previously reported (Heaney et al., 2022; Kondo et al., 2019). Future studies using more granular age groups, such as children under the age of five, may identify potential differential effects by age (Heaney et al., 2022). Among people who are Black, we observed trending positive associations with EDVs for several cardiovascular disease subgroups, namely AMI and dysrhythmia. People who are Black experience environmental health inequities due to a variety of factors, including but not limited to, higher exposure to air pollution, due to discriminatory redlining practices, racial discrimination in the health care system, and barriers to green space access (Jennings and Johnson Gaither, 2015; Lane et al., 2022; Robinson et al., 2023; Yearby et al., 2022). We also observed trending greater positive associations in the risk of EDVs for pneumonia, AURI, and IHD among people who are API. There may be additional disparities within APIs that were masked by the aggregation of data, since the racial/ethnic category of API consists of great diversity in ethnicities, immigration patterns, socioeconomic status, and health outcomes (Budiman and Ruiz, 2021a, 2021b; Yi et al., 2022). Finally, the increases in the risk of asthma EDVs and of CLRD EDVs were trending higher among females compared to males, which is consistent with what has been reported previously (Kondo et al., 2019; Reid et al., 2016b). Future studies with larger numbers of EDVs and more disaggregated data, including specific subgroups within API, could help better detect potential health disparities associated with smoke events among, and within, subgroups for various outcomes.

This study had several strengths. We used wildfire-specific $PM_{2.5}$ concentrations derived from an ensemble machine learning model, which allowed us to hone in on the specific health effects of smoke events. Using ZCTA-level exposures versus monitor data also provided better spatial coverage across the state. In addition, our study covered multiple fires across recent years in California when wildfire smoke exposure has increased and impacted many populations across the state, including urban areas that were not previously affected by large-scale wildfire smoke exposure. Therefore, our findings are likely to be more generalizable to other areas that experience wildfires with similar wildfire smoke composition. We investigated a wide range of disease categories, including those that were rarely examined in previous wildfire smoke studies, such as schizophrenia. Our study also adds to the few studies that have assessed effect modification by race/ethnicity on the health effects of wildfire smoke.

Our study also had several limitations. We used air basins as our unit of analysis to obtain sufficient case counts to conduct stratified analyses. However, there may be more exposure misclassification with aggregation to a larger geographic area. Since we did not include four air basins in Southern California, our results may be less generalizable to that area. We were unable to account for potential underestimation of case counts due to barriers in accessing healthcare or changes in healthcare-seeking behaviors during an active wildfire. We were also unable to investigate health effects for specific sensitive populations that are disproportionately exposed to wildfire smoke, including undocumented agricultural workers, people who are unhoused, and Indigenous peoples, due to insufficient case numbers and lack of information in our data sources (Méndez et al., 2020). In addition to increased exposure to wildfire smoke, Hispanic or Indigenous farmworkers are disproportionately impacted by racism, inadequate labor protections and job insecurity, linguistic isolation, reduced access to health care, and fear of deportation (Méndez et al., 2020). While we identified an increased risk of diabetes EDVs among people who are Hispanic, there are likely additional health inequities that we were unable to detect in our study.

5. Conclusion

We found that smoke events were associated with increased risk of EDVs for all respiratory diseases combined, asthma, CLRD, all cardiovascular diseases combined, and schizophrenia and decreased risk of EDVs for substance use disorders and suicide. There was some indication of effect modification by race/ethnicity, including a higher risk of EDVs for diabetes among people who are Hispanic. With the growing risk of more frequent and more severe wildfires due to climate change, understanding the differential health effects of wildfire smoke and identifying the structural barriers faced by the most susceptible populations is critical. These findings may help inform public health practice to protect populations that are disproportionately impacted by wildfire smoke and to plan for adequate healthcare services during and after smoke events.

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Institutional Review Board statement

Institutional Review Board approval was obtained from California's Committee for the Protection of Human Subjects (project number 12-04-0075).

CRediT authorship contribution statement

Annie I. Chen: Conceptualization, Methodology, Data curation, Investigation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Keita Ebisu:** Methodology, Data curation, Formal analysis, Visualization, Writing – review & editing. **Tarik Benmarhnia:** Methodology, Data curation, Writing – review & editing. **Rupa Basu:** Conceptualization, Methodology, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The health data that were used are confidential. Daily, ZCTA-level wildfire-specific $PM_{2.5}$ concentrations are available here: https://github.com/benmarhnia-lab/Wildfire_PM25_California_ZIP. Air quality and meteorological data from monitors are publicly available as cited in the text. Data from the 2015-2019 5-year American Community Survey may be obtained from the Census Bureau via the tidycensus package in R.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envres.2023.117154.

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Glossary

AMI: acute myocardial infarction
API: non-Hispanic Asian and Pacific Islander
AURI: acute upper respiratory infections
CLRD: chronic lower respiratory disease
EDVs: emergency department visits
IHD: ischemic heart disease
PVD: peripheral vascular disease
TIA: transient ischemic attack